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DERIVATIVES - SWAPS



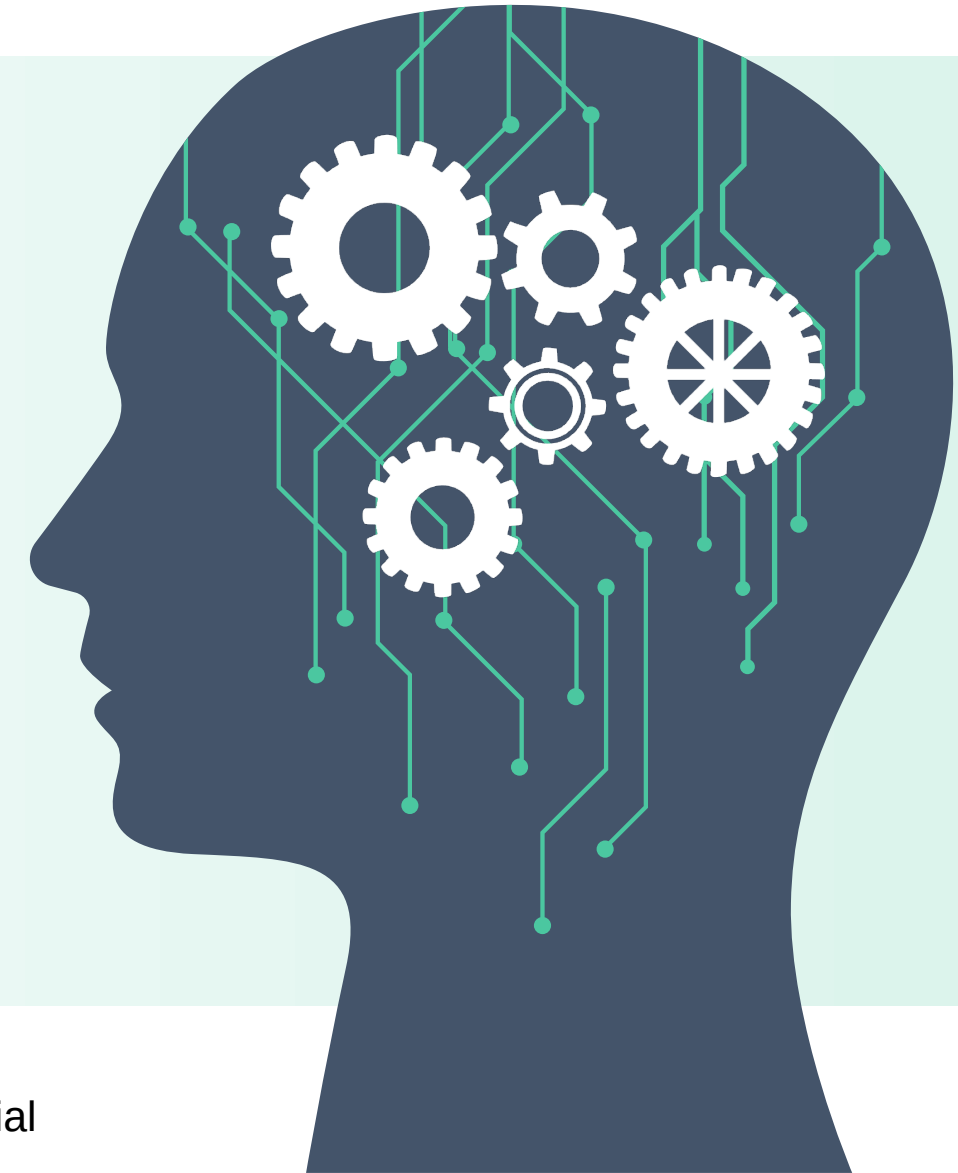
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YOU ARE GOING TO LEARN…

- Introduction to Swaps
- Features & Purpose of Swaps
- Types of Swaps
- Interest Rate Swap
- Equity Swap
- Total Return Swap
- Dividend Swap
- Credit Default Swap



A SWAP AGREEMENT

Swap Agreement



The date when the cash flow has to be paid

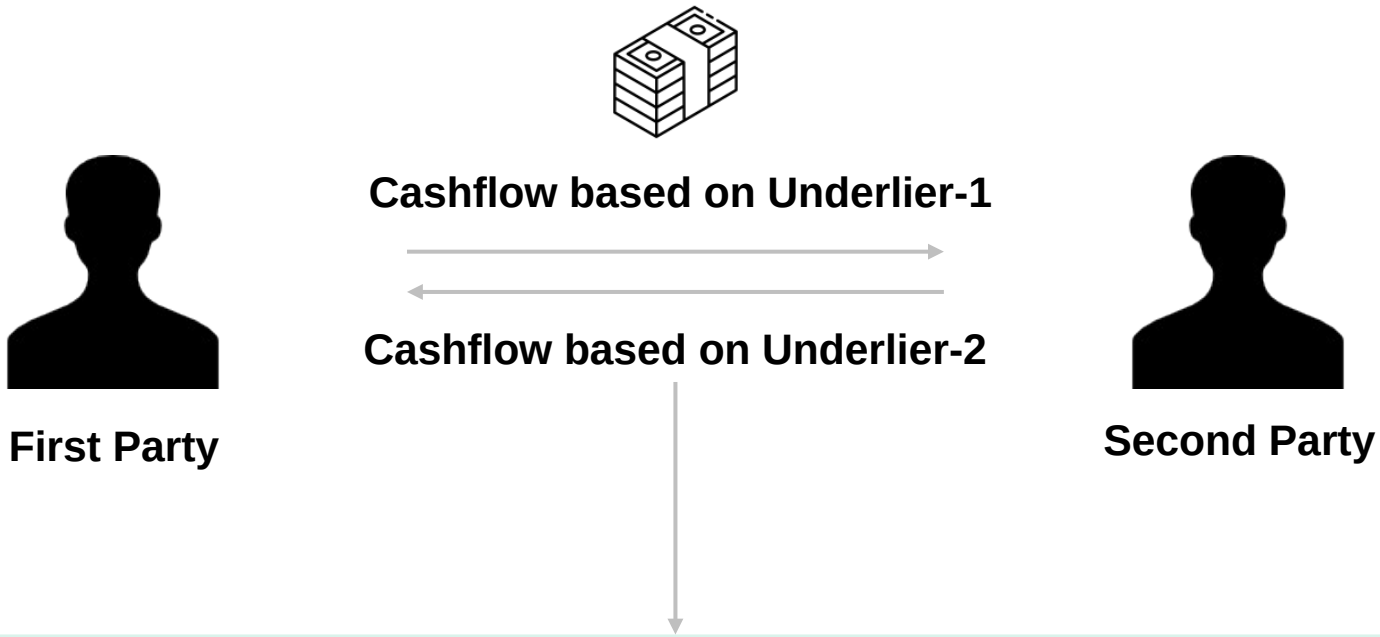
The way the cash flows are accrued and calculated



SWAP is an OTC Derivative Contract wherein the Two Counterparties agrees to exchange the series of cashflows calculated on on two different underlying assets performance on preagreed Notional Amount and this cashflows are exchange on multiple settlement dates over the life of the Swap

UNDERLYERS IN A SWAP AGREEMENT

Two counterparties exchange cash flows based on an underlier.



- Equities
- Commodities
- Credit Risk
- Currencies
- Interest Rate

REASONS FOR TRADING IN SWAPS

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Why do people trade in swaps?



HEDGE RISK

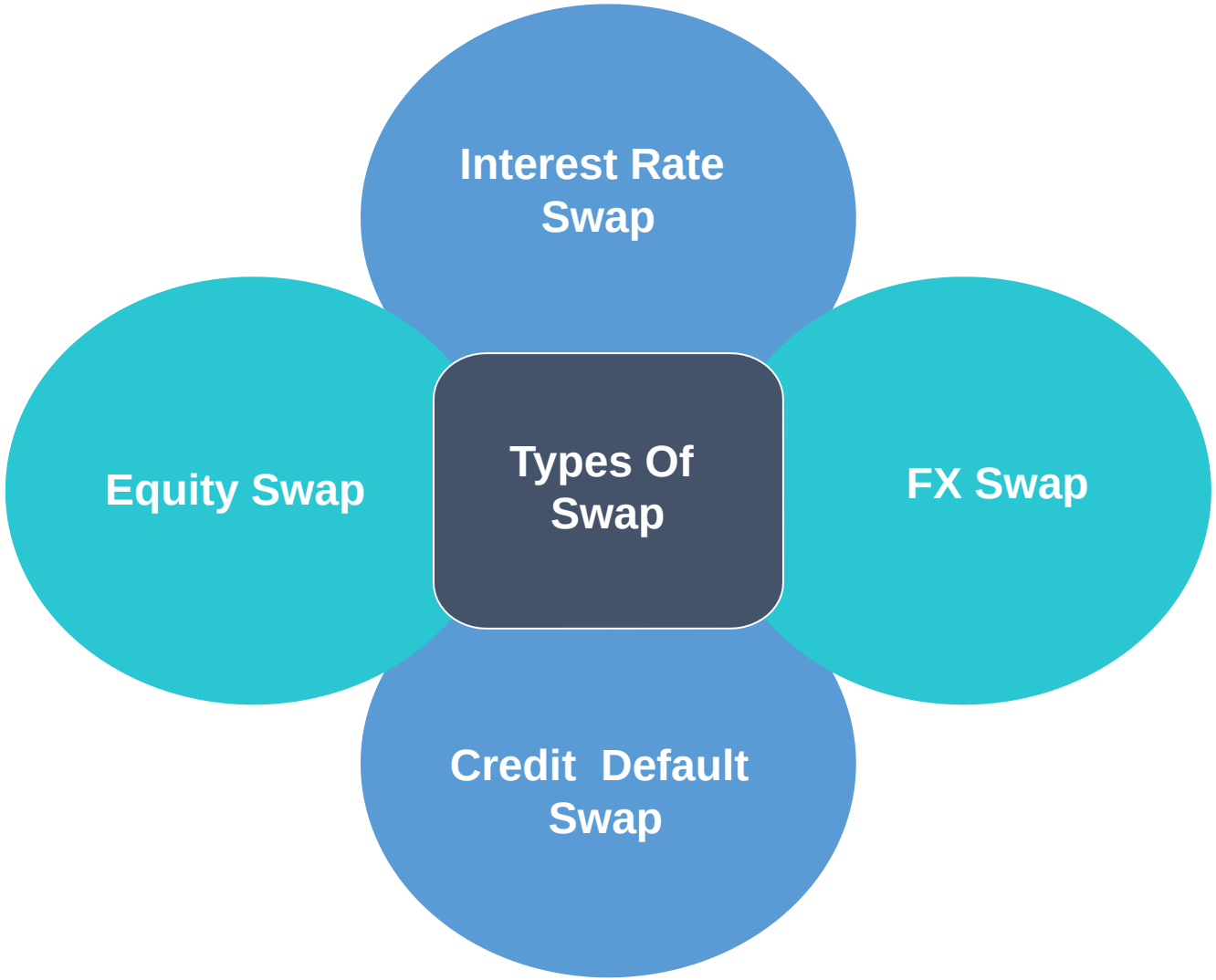


SPECULATION

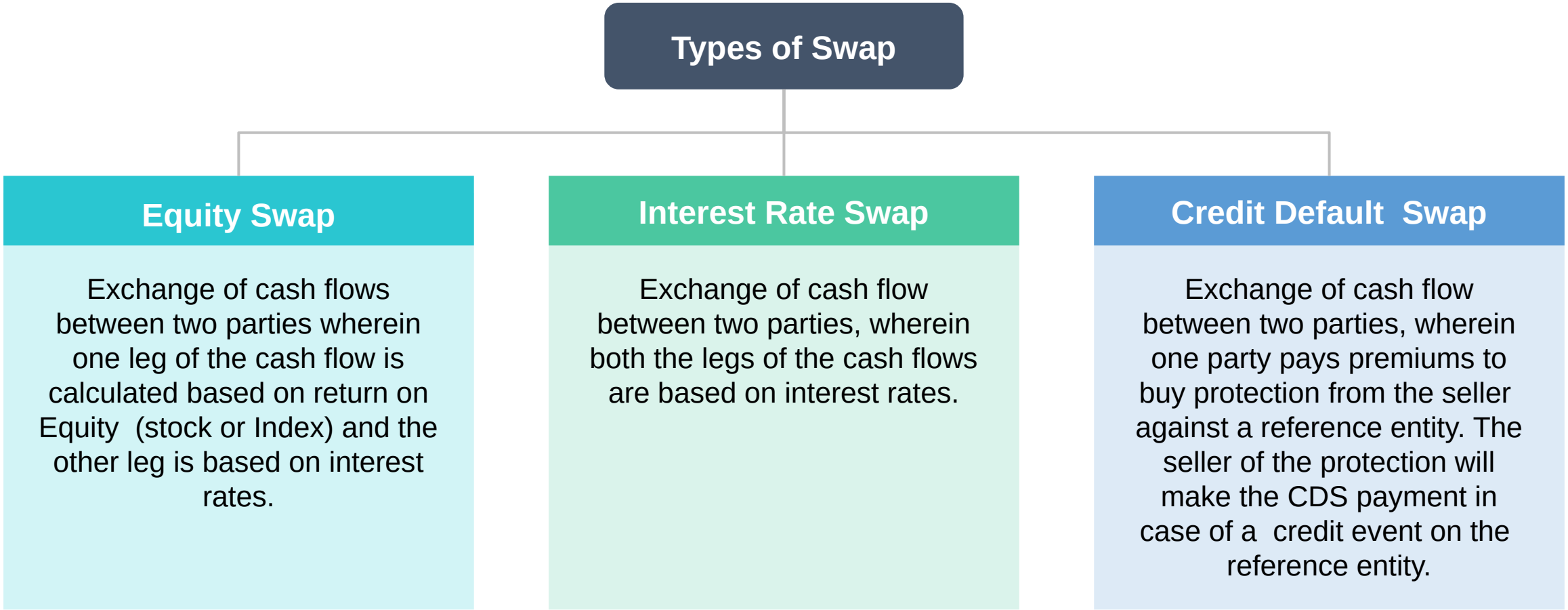


Introduction and Types of Swaps

TYPES OF SWAPS



TYPES OF SWAPS





Interest Rate Swap

INTEREST RATE SWAP (IRS)

An interest rate swap (IRS) is a financial derivative contract between two parties that involves the exchange of interest rate cash flows. In an interest rate swap, one party typically pays a fixed interest rate while the other party pays a variable (floating) interest rate. These swaps are used primarily to manage interest rate risk, reduce borrowing costs, or speculate on interest rate movements.

Types of Interest Rates

01

Fixed Interest Rate

A **fixed interest rate**, also known as a fixed-rate, is an interest rate that remains constant throughout the specified period of a financial transaction or investment. In other words, it does not change over time, regardless of market interest rate fluctuations. Fixed interest rates are commonly used in various financial products, such as loans, bonds, and certain types of investments.

Explanation:

- A fixed interest rate is set at the beginning of a financial arrangement and remains the same for the entire duration of that arrangement.
- This means that the borrower or investor knows exactly how much interest they will pay or earn over the life of the transaction, providing predictability and stability in cash flows.

Floating Rate

02

A floating rate, also known as a variable rate or adjustable rate, is an interest rate that can change periodically based on a reference benchmark or index. Unlike a fixed interest rate, which remains constant over the life of a financial product, a floating rate adjusts to reflect current market conditions. Floating rates are commonly used in various financial products, such as adjustable-rate mortgages, floating-rate bonds, and certain types of loans. Here's a simple explanation and an example of a floating interest rate:

Explanation:

- A floating interest rate is not fixed but rather fluctuates in response to changes in a specified benchmark interest rate or index.
- The rate adjustment typically occurs at predetermined intervals, such as monthly, quarterly, or annually.
- Borrowers or investors with floating-rate products are exposed to changes in market interest rates, which can result in varying interest payments or returns.

USES OF IRS



Portfolio Management



Speculation



Corporate Finance



Risk Management



Rate-locks on Bond Issuance

INTEREST RATE SWAPS: KEY TERMS

Trade Date	Date on which the trade is agreed
Effective Date	Date from which the Swap is Effective
Reset Frequency	How often will the Interest rates be re-calculated
Reset Date	The Date on which the Interest Rates are ‘recalculated’ or ‘readjusted’ to the market rates
Payment Date	Date on which the funds are exchanged
Termination Date	The last reset date of the swap
Term	The time period of a swap
Notional	The amount on which the cash-flows are calculated

IRS

Let’s use this Example of IRS - No need to differentiate between Hedging and Speculation.

In this example, we'll consider a hypothetical IRS between **General Motors** and **IBM Corpn**.

Parties Involved:

- Company General Motors (Fixed-Rate Payer)
- Company IBM Corpn (Floating-Rate Payer)

Terms of the IRS:

- Notional Principal Amount: 10 million $
- Fixed Interest Rate (Company General Motors): 4.5% per annum
- Floating Interest Rate (Company IBM Corpn): 6-Month SOFR* + 2% (Floating Rate Benchmark)
- Payment Frequency: Semi-Annual
- Term of the Swap : 2 Years

*SOFR - Secured Overnight Financing Rate ; an US Benchmark Floating Interest rate

IRS Timeline: 2 years

Let's assume hypothetical 6-Month SOFR Benchmark Interest rates for each half-year and calculate the cash flows for both Company General Motors (Fixed-Rate Payer) and Company IBM Corpn (Floating-Rate Payer) in our 2-year Interest Rate Swap (IRS). For this example, we'll use the following hypothetical 6-Month SOFR rates:

- Year 1 - H1 : 6-Month SOFR = 3.0%
- Year 1 - H2 : 6-Month SOFR = 2.2%
- Year 2 - H1 : 6-Month SOFR = 2.5%
- Year 2 - H2 : 6-Month SOFR = 2.8%

Year 1 - 30th June:

- Fixed Interest Payment (Company Gen Motors): ($10,000,000 * 4.5% / 2) = $225,000 (fixed)
- Floating Interest Payment (Company IBM Corpn): (3.0% + 2%) / 2 = 2.5% * ($10,000,000) = $250,000

Year 1 - 31st Dec:

- Fixed Interest Payment (Company Gen Motors): ($10,000,000 * 4.5% / 2) = $225,000 (fixed)
- Floating Interest Payment (Company IBM Corpn): (2.2% + 2%) / 2 = 2.1% * ($10,000,000) = $210,000

Year 2 - 30th June:

- Fixed Interest Payment (Company Gen Motors): ($10,000,000 * 4.5% / 2) = $225,000 (fixed)
- Floating Interest Payment (Company IBM Corpn): (2.5% + 2%) / 2 = 2.75% * ($10,000,000) = $275,000

Year 2 - 31st Dec:

- Fixed Interest Payment (Company Gen Motors): ($10,000,000 * 4.5% / 2) = $225,000 (fixed)
- Floating Interest Payment (Company IBM Corpn): (2.8% + 2%) / 2 = 2.9% * ($10,000,000) = $290,000

Now, let's calculate the cash flows for each year:

Settlement	Gen Motors Payments	IBM Corpn Payments	Net Settlement Amount
Date of Settlement	**Fixed Interest Rate Payments Payment @4.5% p.a Notional Amt x Int Rate**	**Floating Interest Rate Payments 6m SOFR + 2% Notional Amt x Floating Rate**	**Net Settlement Amount between the Parties**
30th June 20X1	225000 US$	250000 US$	(250000-225000) IBM Corpn Pays 25000 US$ to Gen Motors
31st Dec 20X1	225000 US$	210000 US$	(225000-210000) Gen Motors Pays 15000 US$ to IBM Corpn
30th June 20X2	225000 US$	275000 US$	(275000-225000) IBM Corpn Pays 50000 US$ to Gen Motors
31st Dec 20X2	225000 US$	290000 US$	(290000-225000) IBM Corpn Pays 65000 US$ to Gen Motors

In this table:

- Company GM Motors (Fixed-Rate Payer) pays a fixed interest rate of $225,000 every six months.
- Company IBM Corpn (Floating-Rate Payer) pays a floating interest rate based on the 6-Month SOFR rates plus the 2% spread.

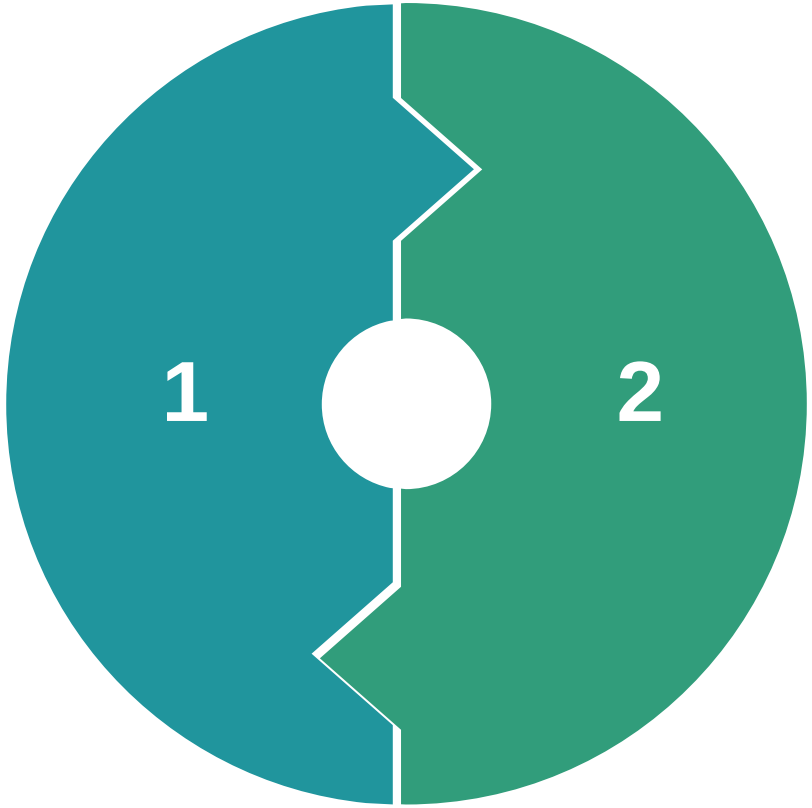
TYPES OF INTEREST RATE SWAP

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Types of IRS

Fixed for floating or ‘vanilla’ swaps

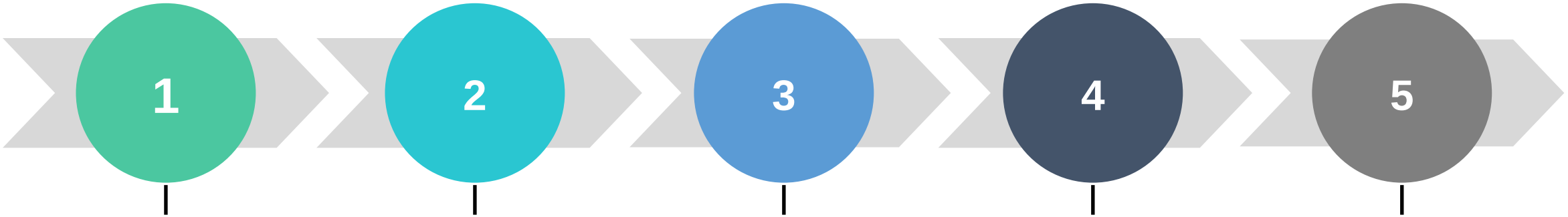
Involves the exchange of a fixed interest payment vs. a floating interest payment



Floating for floating or ‘Basis’ swaps

Involves the exchange of a floating interest payment vs. a floating interest payment

ILLUSTRATION ON TIMELINE OF AN IRS



RISKS ASSOCIATED WITH IRS



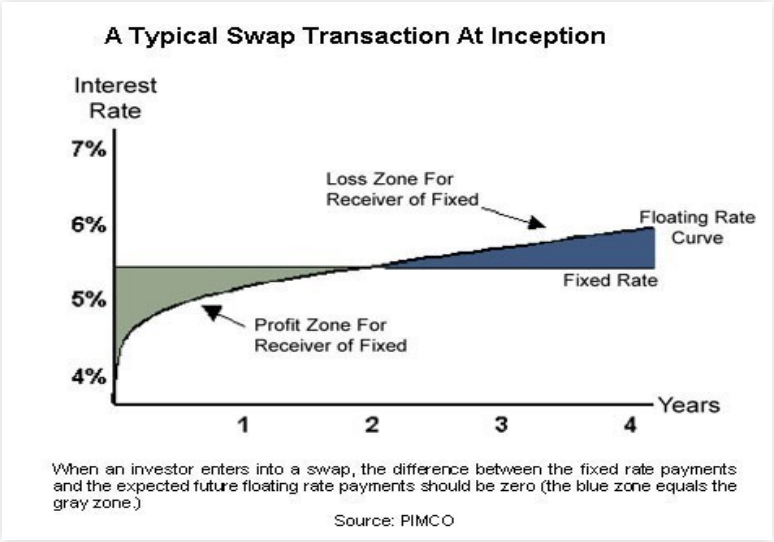
Fixed Interest Rate Receiver

Profit when Interest Rates Fall	Profit when Interest Rates Rise
Loss when Interest Rates Rise	Loss when Interest Rates Fall



Fixed Interest Rate Payer

Primary Risks
Interest Rate Risk
Credit Risk (Counterparty Risk)



Counterparty's Credit Risk
The chance that the other party in the contract will default on its responsibility.



Equity Swap

EQUITY SWAPS

An **Equity Swap** is a financial derivative contract in which two parties agree to exchange the cash flows associated with two different financial instruments, typically related to equities (stocks) or equity indices. The primary purpose of an Equity Swap is to transfer or swap the risks and returns of the underlying assets between the counterparties without actual ownership or transfer of those assets. Equity Swaps are commonly used for various purposes, including hedging, speculation, and portfolio management.

The key components and features of an Equity Swap:

1. Counterparties: Two parties, often referred to as the "fixed leg" and the "floating leg," enter into the swap agreement.

2. Underlying Assets: The swap is based on the performance of underlying assets, which can include individual stocks, stock indices, or baskets of stocks.

EQUITY SWAPS

3. Notional Principal: Like other types of swaps, Equity Swaps involve a notional principal amount. However, this amount is not exchanged between the parties; it serves as the reference point for calculating cash flows.

4. Payment Frequency: Cash flows in Equity Swaps can be structured to occur at various intervals, such as quarterly or annually.

5. Termination Date: The swap agreement specifies the termination date or maturity date of the contract when the final exchange of cash flows occurs.

6. Cash Flows:

- The "**fixed leg**" party typically agrees to pay a fixed interest rate or return to the other party.
- The "**floating leg**" party agrees to pay the returns generated by the underlying assets, which can include dividends, capital gains/losses, or interest income.

EQUITY SWAPS

Equity Swaps can serve several purposes, including:

01

Hedging: Investors can use Equity Swaps to hedge against the risk associated with their equity holdings. For example, if an investor owns a portfolio of stocks and wants to protect against a potential decline in their value, they can enter into an Equity Swap that pays them the return of the portfolio in exchange for a fixed payment.

02

Portfolio Replication: Asset managers and institutional investors can use Equity Swaps to replicate the performance of a specific stock index or benchmark without actually buying all the individual stocks in that index.

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03

Tax Efficiency: Equity Swaps can be structured to achieve specific tax benefits or efficiencies for certain types of investors.

04

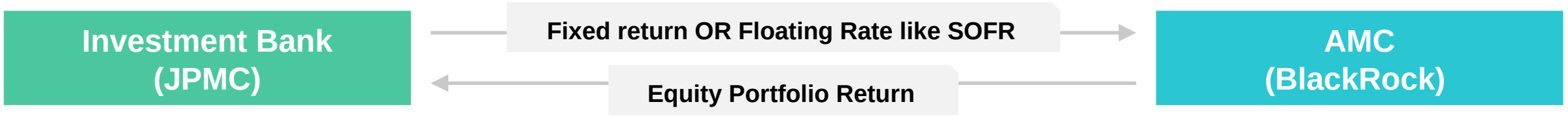
Leverage and Speculation: Traders and investors can use Equity Swaps to gain leveraged exposure to a particular stock or index without purchasing the assets outright. This allows for more significant potential gains (and losses) with a smaller upfront capital requirement.

It's important to note that Equity Swaps can be quite customizable, with various structures and terms to meet the specific needs of the parties involved. However, they also involve counterparty risk, as each party relies on the other to fulfill their payment obligations.

EQUITY SWAPS



- An equity swap contract is a derivative contract between two parties that involves the exchange of one stream (leg) of equity-based cash flows linked to the performance of a stock or an equity index with another stream (leg) of fixed-income cash flows.
- In equity swap contracts, the cash flows are based on a predetermined notional amount. The exchange of cash flows occurs on series of fixed dates.



01 Equity Swap Uses

- Synthetic exposure to equities – no huge investment required.
- Effective hedging tool against volatilities.
- Access to variety of securities.

02 Variations of Equity Swap

- Receive-equity return, pay-fixed
- Receive-equity return, pay-floating
- Receive-equity return, pay-another equity return

TERMINOLOGY ASSOCIATED WITH EQS

Currency	The CCY of the transaction
Direction	Buy means Paying the Interest Rate & Receive Equity Returns Sell means Paying the Equity Performance & Receive Interest Rate
Counterparty	The Counterparty to the transaction
Equity Underlyer	The Equity Asset (Single EQ Stock or EQ Basket or an EQ Index)
Interest Rate	The applicable Interest Rate. Can be Fixed or Floating
Spread	The spread which is implemented on the Floating Interest Rate
Day Count Convention	Counts the no. of days between the two reset dates in order to calculate the payments

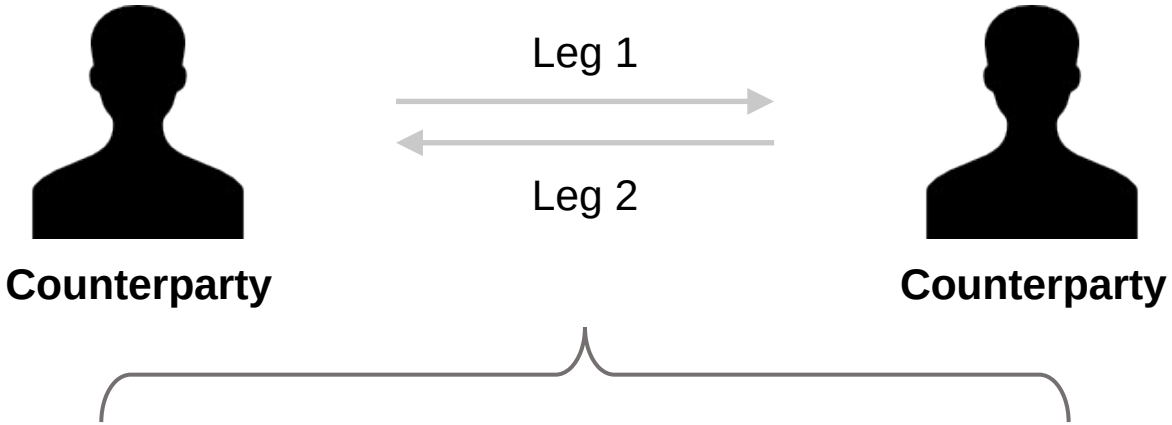
EQUITY SWAPS - TRADE ECONOMIES

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General Terms	Equity Amounts	Floating Rate Amounts	Settlement Terms
• Party A , Party B, Notional Amount, Trade Date, Effective Date, Termination Date • Index • Exchanges • Related Exchanges	• **Equity Amount Payer:** [Party A/B], Equity Amount Receiver [Party A/B], Equity Notional, • **Type of Return:** Price Return/Dividend Return /Total Return, Valuation Dates, Valuation Time	• **Floating Amount Payer:** [Party A/B] Floating Amount, Floating Amount Payer, Payer Currency Amount, Payment Dates • Floating Rate Option, Designated Maturity - Fixing Day - Fixing Time Spread: [Plus/Minus %] • Floating Rate Day Count Fraction • **Reset Dates:** Rate Cut-off Dates , Method of Averaging	• Settlement Currency, Cash Settlement - Applicable

TWO LEGS OF AN EQUITY SWAP

The two cash flows are referred to as "legs" of the swap.



First Leg

- Fixed /Floating Rate
- Interest Leg
- Calculated based on Agreed Notional Amount

Second Leg

- Performance of a Share or Stock Market Index
- Equity Leg
- Calculated based on Agreed Notional Amount

EQUITY SWAPS - EXAMPLE

Terms of the New Equity Swap (Excluding Dividends) : Price Return Swap

- Notional Principal: $1,000,000
- Fixed Interest Rate Payer : Party A ; 5% p.a
- Underlying Stock: CISCO Systems Inc
- Underlying Stock Price Return Payer : Party B
- Stock's Initial Price: $50 per share
- Swap Term: 3 years

Years 1 to 3:

- Party A (Fixed Leg) agrees to pay Party B a fixed interest rate of 5% on the notional principal amount.

- Party B (Floating/Equity Leg) Agrees to Pay Party A the returns based on the capital appreciation or depreciation of CISCO Systems Inc. No dividends are included in this scenario as this SWAP is a Equity Price Return Swap.

EQUITY SWAPS - EXAMPLE....Contd

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Year	Party-A Payment	CISCO EQ Stock	Price Return	Party-B Payment	Net Payment
Settlement Year end	**Fixed Interest Payment @5% p.a Notional Amt x Int Rate**	**Cisco Systems Stock Price on NYSE**	**Equity Stock Price Return %**	**CISCO Equity Stock Capital Gain or Loss Notional Amt x Equity Return**	**Net Settment Amount between the Parties**
Year-0		50			
Year-1	1,000,000 * 5% 50000 US$	55	=(55-50)/50 =10%	=10% * 1,000,000 =100000 US$ Capital Appreciation	(100000-50000) 50000 US$ paid by Party B to Party A
Year-2	1,000,000 * 5% 50000 US$	57	=(57-55)/55 =3.64%	=3.64% * 1,000,000 =36400 US$ Capital Appreciation	(50000-36400) 13600 US$ paid by Party A to Party B
Year-3	1,000,000 * 5% 50000 US$	56	=(56-57)/57 =1.75%	=-1.75% * 1,000,000 -17500 US$ Capital Loss	(50000-(-17500)) 67500 US$ paid by Party A to Party B since the capital loss is also need to be borne by Party A

EQUITY SWAPS - EXAMPLE

- **Party A (Fixed Leg)** pays a fixed interest rate of 5% on the notional principal in each year.
- **Party B (Floating Leg) pays** returns based on the capital appreciation or depreciation of CISCO Systems Inc. No dividends are considered in this example.
- The **"Net Payment (A - B)"** column represents the difference between the fixed interest rate payments made by Party A and the capital appreciation or depreciation of the stock by Party B for each year.

At the end of the 3-year term, Party A's net payments to Party B reflect the fixed interest rate payments made by Party A and the capital appreciation or depreciation of the underlying stock.



Equity Total Return Swap

EQUITY TOTAL RETURN SWAPS (TRS)

A **Total Return Swap (TRS)** is a financial derivative contract between two parties, where one party agrees to pay the total return of a specific asset or basket of assets to the other party in exchange for a predetermined payment. The total return includes all aspects of the asset's performance, such as price appreciation, interest or dividend income, and any other income or gains associated with the asset.

Features of a Total Return Swap:

1. Asset Exposure

TRS allows one party (the "receiver" or "buyer") to gain exposure to the performance of a particular asset, typically without owning it. The other party (the "payer" or "seller") retains ownership of the asset.

2. Total Return

The receiver of the TRS is entitled to the total return on the asset, which comprises capital gains or losses, interest income, dividend income, and any other economic benefits generated by the asset. This makes TRS different from traditional equity swaps, which may only involve capital gains or losses.

TOTAL RETURN SWAPS (EQUITY TRS)

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Here is an example of a TRS

A Total Return Swap (TRS) contract between two parties, Party A and Party B, with a notional principal of $10 million. In this scenario, Party A will receive the total return on an equity, Stock XYZ (Price Return + Dividends), while Party B will receive the fixed Interest Rate of 10% p.a. The TRS has a term of 2 years.

Terms of the TRS:

- Notional Principal: $1,000,000
- Fixed Interest Rate Payer : Party A
- Fixed Interest Rate Agreed : 10% p.a
- Equity Total Return Payer : Party B
- Equity Reference Asset : Stock GOOGLE
- Initial Price of Stock GOOBLE: $100 per share
- Swap Term: 2 years

TOTAL RETURN SWAPS (TRS)

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Year	Party-A Payment	GOOGLE EQ Stock Price Returns		Party-B Payment	Net Payment
Settlement Year end	**Fixed Interest Payment @10% p.a Notional Amt x Int Rate**	**GOOGLE Stock Price & Dividend on NYSE**	**Equity Total Return Stock Price Return %**	**GOOGLE Equity Stock Total Return Notional Amt x Equity Return**	**Net Settlement Amount Between the Counter Parties**
Year-0		100			
Year-1	1,000,000 * 10% =100000 US$	115 $/Share Dividend = 2$ per share	=(115-100+2)/100 =17%	=17% * 1,000,000 =170000 US$ Capital Appreciation	(170000-100000) 70000 US$ paid by Party B to Party A
Year-2	1,000,000 * 10% =100000 US$	120 $/Share Dividend = 3$ per share	=(120-115+3)/115 6.96%	=6.96% * 1,000,000 =69600 US$ Capital Appreciation	(100000-69600) 30400 US$ paid by Party A to Party B
Year-3	1,000,000 * 5% =100000 US$	108 $/Share Dividend = 1$/share	=(108-120+1)/120 =-9.17%	= -9.17% * 1,000,000 - 91700 US$ Capital Loss	(100000-(-91700)) 191700 US$ paid by Party A to Party B since the total equity return loss is also need to be borne by Party A

TOTAL RETURN SWAPS (TRS)

In this example: Years-1 to Year-3

Party A (Total Equity Return Receiver)

- **Party A (Total Equity Return Receiver**) receives the total return on Stock GOOGLE, which includes capital appreciation (the difference in stock price from the previous year) & dividend income.

Party B (Fixed Interest Rate Receiver)

- **Party B (Fixed Interest Rate Receiver)** receives the Fixed Interest Rate, which is assumed to be 10% of the notional principal annually.



Credit Default Swap

CREDIT DEFAULT SWAPS

A **Credit Default Swap (CDS)** is a financial derivative contract that allows one party, known as the protection buyer, to transfer the credit risk of a specific debt obligation to another party, known as the protection seller. The primary purpose of a CDS is to provide insurance or protection against the possibility of a credit event, such as a default, bankruptcy, or a significant credit downgrade, occurring with the underlying debt instrument.

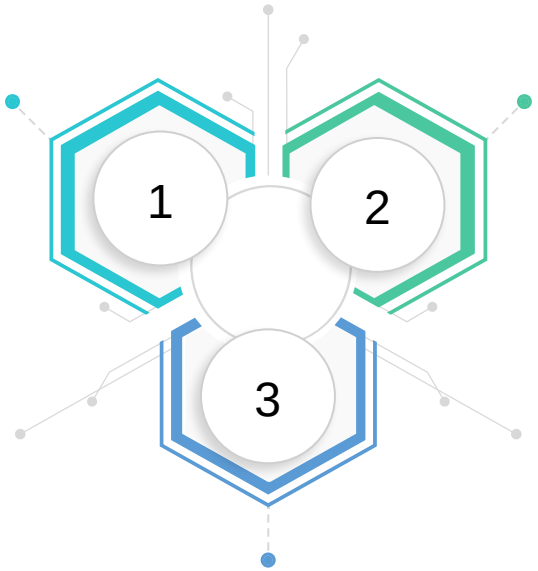


CREDIT DEFAULT SWAPS

Here are the key features and components of a Credit Default Swap:

Reference Obligation

The CDS contract identifies a reference obligation, which is typically a specific bond or debt instrument. The protection buyer owns or holds exposure to this reference obligation.



Protection Buyer

This party seeks to protect itself from the credit risk associated with the reference obligation. It pays regular premiums (similar to insurance premiums) to the protection seller.

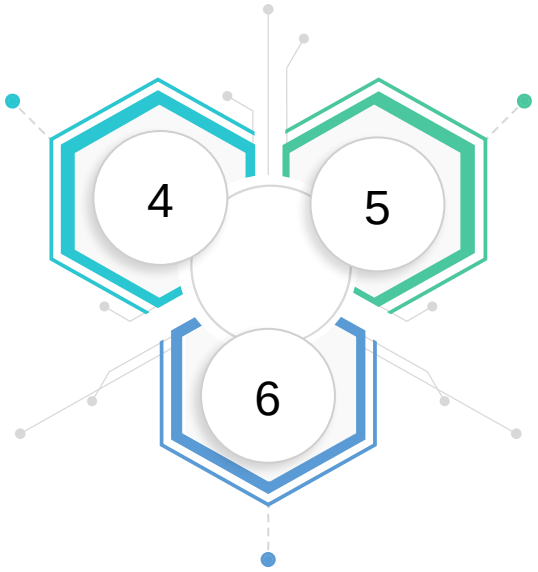
Protection Seller

The protection seller agrees to compensate the protection buyer in the event of a credit event related to the reference obligation. In exchange for this protection, the protection seller receives premium payments from the protection buyer.

CREDIT DEFAULT SWAPS

Notional Amount

The notional amount represents the face value of the reference obligation or the amount of debt being protected. It is not exchanged between the parties and serves as the basis for calculating the payments.



Credit Events

CDS contracts specify what constitutes a credit event. Common credit events include default, bankruptcy, failure to pay, or a significant credit downgrade of the reference obligation. When a credit event occurs, the protection seller must make a payment to the protection buyer.

Premiums

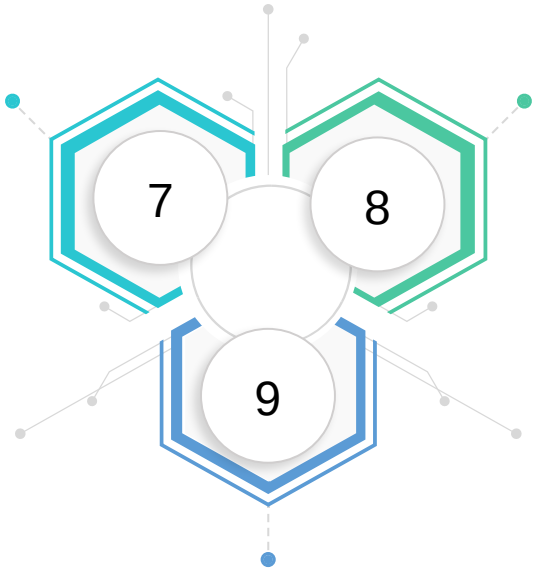
The protection buyer pays periodic premiums to the protection seller. These premiums are similar to insurance premiums and compensate the protection seller for assuming the credit risk. The premium amount is typically a percentage of the notional amount.

CREDIT DEFAULT SWAPS

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Maturity Date

CDS contracts have a defined maturity date, which is when the contract expires. The buyer's credit risk protection ends on the maturity date.



Settlement Terms

CDS contracts can be physically settled or cash settled, depending on the terms of the contract. In physical settlement is when the protection seller pays the buyer the par value of the debt obligation, and the buyer gives the seller the debt obligation in return.In a cash settlement, the protection seller pays the protection buyer the difference between the notional amount and the recovery value of the reference obligation.

Counterparty Risk

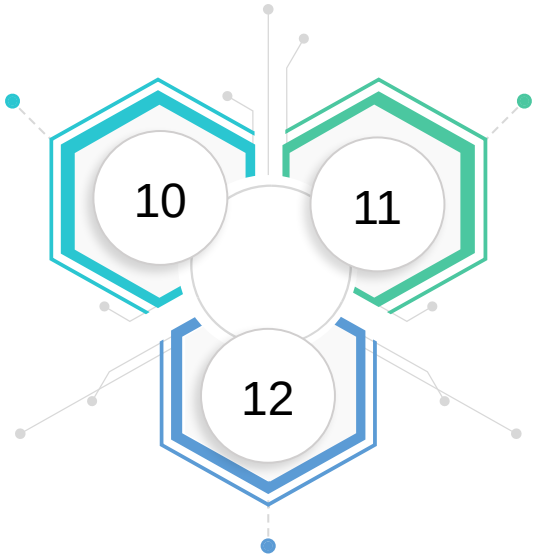
CDS contracts involve counterparty risk, as the protection buyer relies on the protection seller to fulfill their payment obligations in the event of a credit event. Parties may require collateral or use credit support annexes (CSAs) to mitigate this risk.

CREDIT DEFAULT SWAPS

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Trading and Liquidity

CDS contracts are actively traded in financial markets, and their liquidity can vary depending on the reference obligation and market conditions.



Credit Risk Transfer

CDS contracts provide a mechanism for transferring credit risk without the need to buy or sell the underlying debt security. They are often used for hedging credit exposures or speculating on credit events.

Regulation

The regulation of CDS markets and market participants varies by jurisdiction. Regulatory authorities have sought to increase transparency and reduce risks in CDS trading following the 2008 financial crisis.

CDS contracts are versatile financial instruments used by a range of market participants, including banks, hedge funds, corporations, and institutional investors, to manage and mitigate credit risk. They are a key tool for credit risk management and can be an important component of financial risk management strategies.

CREDIT EVENTS

Credit events are specific occurrences that trigger obligations in a Credit Default Swap (CDS) contract. These events typically represent adverse developments related to the creditworthiness of the reference obligation (the underlying debt instrument).



CREDIT EVENTS

The specific types of credit events can vary depending on the terms of the CDS contract, but common credit events include:

Default:
Default is one of the most common credit events in CDS contracts. It occurs when the issuer of the reference obligation fails to make a scheduled interest or principal payment on the debt, violating the terms of the bond or loan agreement.

1

Bankruptcy:
Bankruptcy is another significant credit event. It happens when the issuer files for bankruptcy protection, indicating its inability to meet its financial obligations. Bankruptcy can result in a significant loss for bondholders.

2

Failure to Pay:
This event occurs when the issuer fails to make a scheduled payment of principal or interest on the reference obligation, regardless of whether a formal default or bankruptcy has occurred. It is a broader event than default, encompassing situations where payment obligations are not met.

3

CREDIT EVENTS

The specific types of credit events can vary depending on the terms of the CDS contract, but common credit events include:

Credit Rating Downgrade:
A credit rating downgrade event is triggered when a recognized credit rating agency (e.g., Moody's, S&P, Fitch) downgrades the credit rating of the reference obligation to a level specified in the CDS contract. The downgrade must typically meet certain criteria, such as a minimum rating threshold.

4

Restructuring:
Restructuring events involve changes to the terms of the reference obligation that are adverse to the bondholders. This can include changes to the maturity, interest rate, or principal amount of the debt. Restructuring events can be complex and are typically defined in detail within the CDS contract.

5

Acceleration:
Acceleration events occur when the terms of the reference obligation are altered in a way that accelerates the maturity of the debt. This often happens when a default or bankruptcy occurs, and bondholders demand full repayment.

6

CREDIT EVENTS

The specific types of credit events can vary depending on the terms of the CDS contract, but common credit events include:

Moratorium:
A moratorium event is declared when the issuer of the reference obligation imposes a temporary halt or delay on payments to bondholders, typically during a financial crisis or when seeking to negotiate new terms.

7

Repudiation of Obligations:
This event occurs when the issuer explicitly refuses to acknowledge or honor its debt obligations, even if payment has not yet been due.

8

Cessation of Payments:
This event is triggered when the issuer stops making payments on its debt obligations, which may or may not be associated with a formal default.

9

CREDIT EVENTS

- It's important to note that the specific credit events covered in a CDS contract can vary based on the contract's terms and negotiation between the parties involved. Additionally, the terms and definitions of these events are usually detailed in the contract's documentation and may incorporate criteria from recognized credit rating agencies.

- The occurrence of a credit event in a CDS contract typically leads to a payment from the protection seller to the protection buyer, providing compensation for the credit risk associated with the reference obligation. The precise terms and triggers for these events should be carefully reviewed within the CDS contract to ensure clarity and understanding.

BANKRUPTCY: EXAMPLE

15th September 2008

$639 bn
In Assets

$619 bn
In Debt

LEHMAN BROTHERS

Largest victim of the subprime mortgage crisis

$10 trillion
Erosion in Global Market Capitalization

RESTRUCTURING: EXAMPLE



$600 mn
Compensation

$600 mn
Compensation

$600 mn
Compensation

DERIVATIVES - CREDIT DEFAULT SWAPS - EXAMPLE

1

Terms of the CDS:

- At the beginning of Year-1, Party A purchases a CDS Derivative contract from Party B as protection against the possibility of default by Company XYZ's issued Bonds.
- Notional Principal: $10 million
- Counter Parties : Party A & Party B
- CDS Buyer : Party A
- CDS Seller : Party B
- Reference Entity: Company XYZ Ltd
- Reference Entity Obligation: XYZ' Ltd issued Unsecured Bonds
- Reference Entity Credit Rating : AAA+
- Premium Payment Frequency: Quarterly upfront
- Term/Maturity of CDS: 3 years
- Survival Rate: 95%; Probability of Survival during the CDS Term
- Recovery Rate: 40%; The estimation of recovery amount in the event of default
- CDS Premium : 2% Per Annum but payable Quarterly

DERIVATIVES - CREDIT DEFAULT SWAPS - EXAMPLE

2

Premium Calculation & Payment:

- The survival rate is 95%, meaning there is a 95% probability that Company XYZ Ltd will not Default during the CDS term i.e. 5% probability of Default by the Reference Entity Company XYZ Ltd.
- CDS Premium depends on the credit risk associated with the Reference Entity & Reference Obligation i.e Probability of Default & Recovery Rate
- In this example, the premium rate is set at 200 basis points (2%) per annum, payable quarterly.
- Premium Payment per quarter = (Notional Principal * Premium Rate) / 4

Year-1; Quarter-1:

- At the beginning of Quarter-1 of Year-1, The premium for Q1 is calculated as follows:
- The premium for Quarter-1 of Year-1 is calculated as follows:
- Premium Payment = ($10,000,000 * 2% / 4) = $50,000 per Quarter
- Party A pays $50,000 to Party B as the premium for Quarter-1.

DERIVATIVES - CREDIT DEFAULT SWAPS - EXAMPLE

2

Premium Calculation & Payment:

Year-1; Quarter-2:

- At the beginning of Quarter-2 of Year-1, The premium for Q2 is calculated as follows:
- Premium Payment = ($10,000,000 * 2% / 4) = $50,000
- Party A pays $50,000 to Party B as the premium for Q2.

Year-1; Quarter-3:

- At the beginning of Quarter-3 of Year-1, The premium for Q3 is calculated as follows:
- Premium Payment = ($10,000,000 * 2% / 4) = $50,000
- Party A pays $50,000 to Party B as the premium for Q2.

DERIVATIVES - CREDIT DEFAULT SWAPS - EXAMPLE

3

Credit Event Occurrence:

- In Quarter-3, Company XYZ encounters financial difficulties and defaults on its bonds. This event triggers the CDS contract.
- The market value of the bonds is equal to 40% recovery rate, indicating that in the event of default, the bondholders can expect to recover 40% of the face value of the bonds hence there is a financial loss of 60% of Face Value
- Party A, as the protection buyer, experiences a credit event (default) and is entitled to a payout from Party B, the protection seller.

The payout to Party A is calculated as follows: Payout = (Notional Principal *(1-Recovery Rate))= ($10,000,000 * 60%) = $6,000,000

Party B, the protection seller, must pay $6,000,000 to Party A as compensation for the credit event (Company XYZ's default).

DERIVATIVES - CREDIT DEFAULT SWAPS - EXAMPLE

3

Credit Event Occurrence:

Q4 onwards :

- Since the CDS contract has already triggered due to default, no further premium payments are made.

In this example:

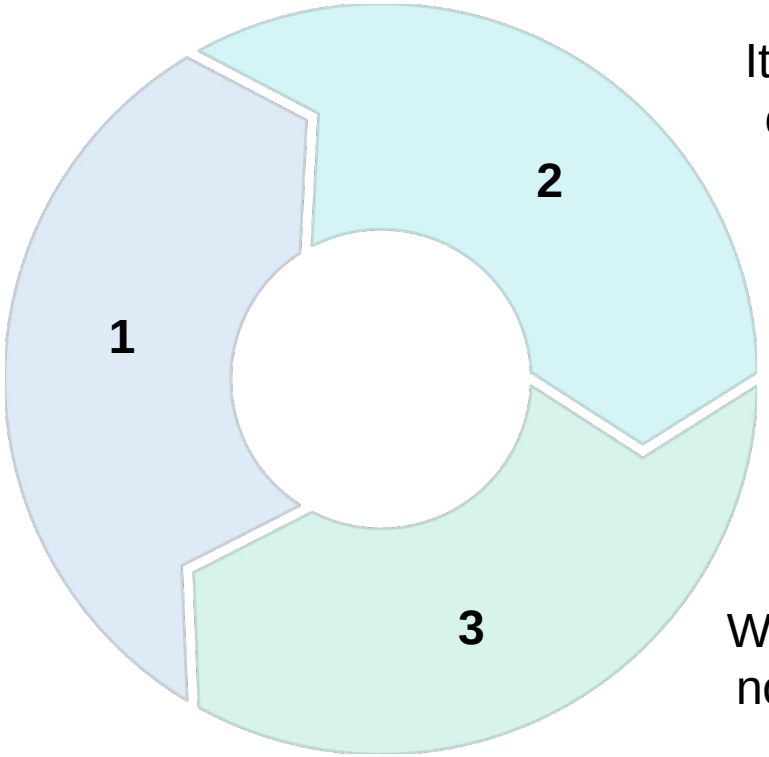
- Party A, the protection buyer, paid premiums to Party B, the protection seller, over the first three quarters of Year-1.
- When Company XYZ defaulted in Quarter-3, the credit event occurred, and Party A received a payout of $6,000,000 from Party B, which is based on the recovery rate of 40% as the Loss is 60%.

TYPES OF CREDIT DEFAULT SWAP

IMARTICUS LEARNING

Single Name CDS

It considers the debt obligation of only one company as its reference obligation



Basket CDS

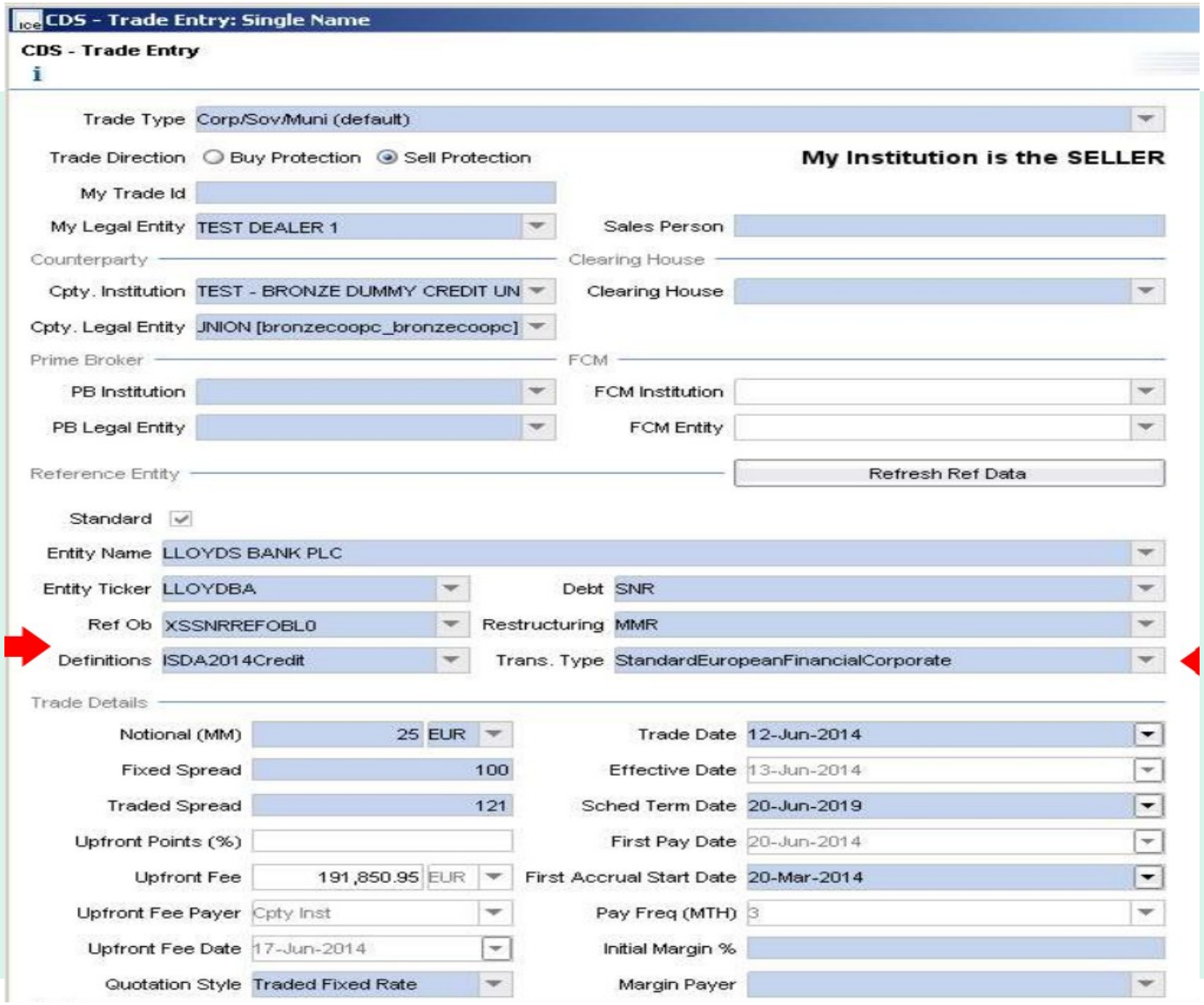
It considers the debt obligations of more than one company as its reference obligation.

Naked CDS

When the buyer of the protection does not hold the debt obligation. Typically, CDS Buyer bets on a reference obligation, which they does not owns, speculation trade.

DERIVATIVES - SWAPS

Credit Default Swaps – Trade Economics



Its time to play KAHOOT Quiz



Click here to start the quiz



IMARTICUS LEARNING

THANK YOU !!!

Contact us today!
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